

1 Entry of Short-Term Malfunctions

Utilization

To improve the overview, e.g. for the status protocol one status per machine may be defined as short-term disturbance. This status is used as “container” for *unconfirmed* statuses, which were only available for a specific (short) time while monitoring the machine.

If a downtime is automatically detected at the terminal and the machine goes back into production automatically, the system verifies if the duration of this disturbance is shorter than the duration configured for this machine as short-term disturbances.

If it is, then the disturbance showing no reasons up until now is set with a reason using the status configured at the machine as the status for "short-term disturbances" (MDE configuration --> Administration --> Waiting period for short-term disturbances).

Such automatic status postings (reasons) for short-term disturbances are displayed within HYDRA in exactly the same manner as if they had been given a reason by an operator.

Configuration of short-term status - status assignment

Within status assignment 1 status (at most) is defined with the control indicator “short-term disturbance”.

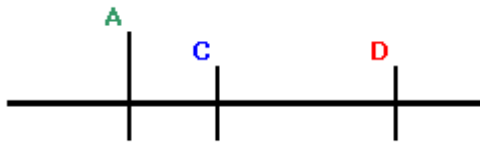
Configuration of the short-term status - configuration of machines/workplaces

The time until which a disturbance is still interpreted as short-term disturbance is defined within the configuration of machines/workplaces. If the time configured for short-term disturbances is 0 the function is deactivated.

Processing - scenarios

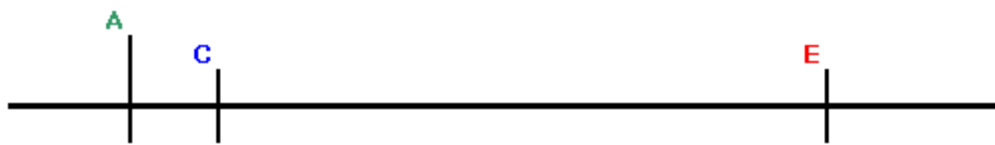
Scenario 1: Machine monitoring, manual disturbance message before short-term disturbance expires

- Time A: beginning of malfunction; terminal does not receive signal
- Time C: expiry of minimum cycle time including cycle extension according to HYDRA-MDE machine configuration → terminal switches to “not assigned” (30000)
- Time D: manual confirmation of disturbance reason → disturbance reason is assigned to time A



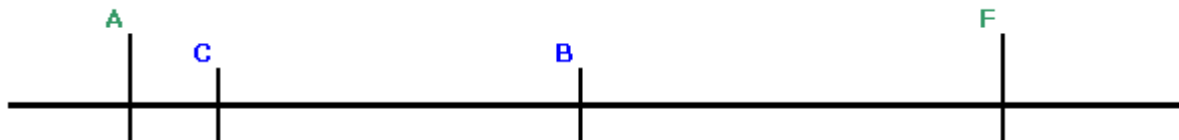
Scenario 2: Machine monitoring; manual disturbance message after short-term disturbance has expired

- Time A: beginning of malfunction; terminal does not receive signal
- Time C: Expiry of minimum cycle time including cycle extension according to HYDRA-MDE machine configuration → terminal switches to "not assigned" (30000)
- Time E: manual confirmation of disturbance reason → disturbance reason is assigned to time A



Scenario 3: Machine monitoring; no manual disturbance message

- Time A: beginning of malfunction; terminal does not receive signal
- Time C: Expiry of minimum cycle time including cycle extension according to HYDRA-MDE machine configuration → terminal switches to "not assigned" (30000)
- Time B: Expiry of "short-term disturbance time"
- Time F: beginning of production → the time between A and F is posted to "not assigned" (→ general malfunction).



Scenario 4: Machine monitoring; beginning of production before short-term disturbance expires

- Time A: beginning of malfunction; terminal does not receive signal
- Time C: Expiry of minimum cycle time including cycle extension according to HYDRA-MDE machine configuration → terminal switches to "not assigned" (30000)
- Time B: Expiry of "short-term disturbance time"
- Time G: Signal from machine → the time between A and G is posted to "short-term disturbance reason".



Please note

In connection with the shift automatic option it might be the case that a short-term disturbance is generated when shifts are changed if the time between detection of the malfunction (status 30000) and the end of the shift is less than the “short-term disturbance time” defined. A short-term disturbance is also generated when the period of time between the beginning of the shift and restarting of production is less than the “short-term disturbance time” defined. This is also the case, when the sum of both periods (before and after the change of shifts) is greater than the short-term disturbance time configured.